

Project Alexandria: Knowledge Units with guarded back-translation

20 disjoint papers · 200 frozen multiple-choice questions per condition · 28 conditions / 5,600 QA slots

The same Qwen2.5-7B answerer uses each context. Generation has thinking disabled. Original failed generations remain wrong. English → German → English repair targets copied descriptive wording; graph identifiers, names, targets and metadata remain intact. No retraining or QA-guided repair selection.

QA comparison: unchanged raw KUs versus guarded paraphrasing

KU generator	Source words	Raw QA	Guarded BT QA	Change	Paired 95% interval	Invalid
Qwen 3.8 27B FP8 · teacher	500	88.5% 177/200	90.0% 180/200	+1.5 pp	+0.0 to +3.0 pp	0
Gemma 4 E4B IT · no adapter	500	75.5% 151/200	74.5% 149/200	-1.0 pp	-3.5 to +1.5 pp	10
Gemma 4 E4B IT · KU LoRA rank 128	500	82.5% 165/200	82.5% 165/200	+0.0 pp	-1.5 to +1.5 pp	10
Gemma 4 12B IT · no adapter	500	75.0% 150/200	75.0% 150/200	+0.0 pp	-2.0 to +2.0 pp	10
Gemma 4 12B IT · KU LoRA rank 128	500	89.0% 178/200	88.5% 177/200	-0.5 pp	-1.5 to +0.0 pp	0
Qwen 3.8 27B FP8 · teacher	1000	88.0% 176/200	87.5% 175/200	-0.5 pp	-2.5 to +1.0 pp	0
Gemma 4 E4B IT · no adapter	1000	66.5% 133/200	66.5% 133/200	+0.0 pp	-3.0 to +3.5 pp	0
Gemma 4 E4B IT · KU LoRA rank 128	1000	79.5% 159/200	78.0% 156/200	-1.5 pp	-3.5 to +0.5 pp	0
Gemma 4 12B IT · no adapter	1000	64.5% 129/200	64.5% 129/200	+0.0 pp	-3.0 to +3.0 pp	10
Gemma 4 12B IT · KU LoRA rank 128	1000	84.5% 169/200	84.0% 168/200	-0.5 pp	-2.0 to +1.0 pp	0

Paired intervals resample whole papers in 10,000 bootstrap draws. An interval including zero does not establish a reliable accuracy change. These questions were teacher-authored and automatically source-validated, not independently human-curated.

Full-paper, no-context and summary controls

Context	QA accuracy	Correct	95% interval	Invalid
Full original paper / ground-truth context	91.0%	182/200	87.5–94.5%	0
No paper context	43.0%	86/200	38.0–48.0%	3
Qwen27 summary	81.0%	162/200	75.5–86.5%	0
Gemma E4B base summary	64.0%	128/200	51.5–75.0%	20
Gemma E4B KU LoRA used for summaries	68.0%	136/200	54.5–80.0%	20
Gemma12 base summary	73.5%	147/200	67.0–79.5%	0
Gemma12 KU LoRA used for summaries	80.5%	161/200	74.0–86.5%	0
Gemma12 earlier summary-trained LoRA rank128	89.0%	178/200	83.5–93.5%	0

Copying, fidelity and measured compute

The primary audit counts all factual KU strings separately, preserving the previous benchmark's definition. Document title/author metadata are separate. The whole source is used: ≤ 5 copied normalized whitespace words passes, exactly six is borderline, ≥ 7 is flagged. Reasoning, prompts and questions are excluded.

Generator	Words	Raw coverage ≥ 6	BT coverage ≥ 6	BT max run	BT ≤ 5 pass	BT ≥ 7 flag
Qwen 3.8 27B FP8 · teacher	500	10.97%	7.85%	31	0/20	20/20
Gemma 4 E4B IT · no adapter	500	22.51%	19.07%	37	0/20	20/20
Gemma 4 E4B IT · KU LoRA rank 128	500	14.80%	11.16%	37	0/20	20/20
Gemma 4 12B IT · no adapter	500	8.16%	6.54%	20	0/20	19/20
Gemma 4 12B IT · KU LoRA rank 128	500	13.47%	10.18%	35	0/20	20/20
Qwen 3.8 27B FP8 · teacher	1000	10.32%	7.56%	22	0/20	20/20
Gemma 4 E4B IT · no adapter	1000	21.27%	18.25%	41	0/20	20/20
Gemma 4 E4B IT · KU LoRA rank 128	1000	14.19%	9.96%	26	0/20	20/20
Gemma 4 12B IT · no adapter	1000	8.35%	6.30%	15	1/20	16/20
Gemma 4 12B IT · KU LoRA rank 128	1000	11.18%	8.55%	37	0/20	20/20

Repair inserted 1,123 windows in 190 / 200 KU document versions. Maximum two round-trips. Rejected edits retain the original field. Acceptance requires bidirectional NLI ≥ 0.9 , unchanged number/unit/formula signatures and no copy run > 5 inside the candidate window.

Raw candidates produced 315 conservative critical-value flags; guarded final documents have 0 retained critical-value flags. This is an automated signature/NLI result rather than independent human verification of every claim.

Active translation: 17.50 seconds / 0.00486 GPU-hours. Full experiment: 0.4500 reserved GPU-hours on four GH200 GPUs, including startup, semantic checks, QA and idle roles. These costs cover the 200 cached KU document versions, not the original extraction or training.

Models, profiles and complete examples

Gemma12 KU adapter: <https://huggingface.co/laion/Alexandria-Gemma-4-12B-it-Qwen27-Knowledge-Units-LoRA-r128>

Gemma E4B KU adapter: <https://huggingface.co/laion/Alexandria-Gemma-4-E4B-it-Qwen27-Knowledge-Units-LoRA-r128>

Qwen teacher: <https://huggingface.co/Qwen/Qwen3.8-27B-FP8>

Each KU adapter was trained at rank128, one epoch, on 391 Qwen27 completions from 97 papers using 1,000-word chunks. 500 and 1,000 words are two inference profiles for the same fixed weights, not separately trained adapters.

Interactive static reader: <https://projects.laion.ai/project-alexandria/ku-examples/>

Attribution, style and sequential context

The paper defines entities, attributes, relationships, contextual notes and sentence MinHash. The current repository sequential prompt supplies up to ten earlier KUs to stabilize naming, but supplies no future text. Its context_summary describes the target under the current schema; it does not guarantee a separate retelling of preceding paragraphs.

Source-registry title and authors are shown for attribution in the reader. The original document author field was not populated, although some entities carry author attributes. There is no dedicated style field. Reader links and adjacent summaries show what comes before/after without adding future facts to model outputs or the QA context.

Condition	Title /20	Author field /20	Author attrs /20	Style attrs /20	Context / KU	MinHash / KU
Qwen 3.8 27B FP8 · teacher · 500	20	0	2	0	163/163	163/163
Gemma 4 E4B IT · no adapter · 500	20	0	0	0	162/162	162/162
Gemma 4 E4B IT · KU LoRA rank 128 · 500	20	0	3	0	162/162	162/162
Gemma 4 12B IT · no adapter · 500	20	0	0	0	162/162	162/162
Gemma 4 12B IT · KU LoRA rank 128 · 500	20	0	6	1	163/163	163/163
Qwen 3.8 27B FP8 · teacher · 1000	20	0	2	0	85/85	85/85
Gemma 4 E4B IT · no adapter · 1000	20	0	0	0	85/85	85/85
Gemma 4 E4B IT · KU LoRA rank 128 · 1000	20	0	3	0	85/85	85/85
Gemma 4 12B IT · no adapter · 1000	20	0	0	0	84/84	84/84
Gemma 4 12B IT · KU LoRA rank 128 · 1000	20	0	7	0	85/85	85/85

Five example papers

Adiabatic quantum search with atoms in a cavity driven by lasers

Evidence for contact delocalization in atomic scale friction

THE 2-ADIC VALUATION OF STIRLING NUMBERS

A six-month periodic fasting reduces microalbuminuria and improves metabolic control in patients with type 2 diabetes and diabetic nephropathy: a randomized controlled study

Intact finger representation within primary sensorimotor cortex of Musician's Dystonia